

Magnetism .

Origin of Magnetism :-

- The electron revolves around the nucleus in the circular orbit, that orbit can be considered as current carrying loop.
- Due to the current carrying loop, the atom shows magnetic dipole moment.
- The magnetic dipole moment is given by -

$$M = I \times A \quad \text{--- (1)}$$

where, I : current due to charge of e^- .

A : Area of orbit.

- If e = charge of electron .

r = radius of orbit .

T = period of revolution of e^- .

v = Speed of e^- .

eqⁿ (1) can be written as -

$$M = \frac{e}{T} \cdot \pi r^2 \quad \text{--- (2)}$$

$$\text{but } v = \frac{2\pi r}{T} \rightarrow T = \frac{2\pi r}{v} \quad \text{--- (3)}$$

substitute eqⁿ (3) in (2) .

$$M = \frac{ev}{2\pi r} \cdot \pi r^2 \quad \text{--- (4)}$$

$$M = \frac{evr}{2}$$

multiply & divide by m_e , where m_e is mass of electron .

$$M = \frac{evr}{2} \times \frac{m_e}{m_e}$$

$$= \frac{e}{2m_e} (vr m_e)$$

but, $m_e v r = L = \text{angular momentum}$.

$$M = \frac{e}{2m_e} (\vec{L}) \quad \text{--- (5)}$$

The above eqⁿ shows that the angular momentum of e^- is responsible for magnetic dipole moment. (origin of magnetism).

- According to Bohr postulate -

$$\vec{L} = n \cdot \left(\frac{h}{2\pi} \right) \quad \text{--- (6)}$$

Substitute eqⁿ (6) in (5) -

$$M = n \left(\frac{eh}{4\pi m_e} \right)$$

for $n=1 \rightarrow \text{first orbit}$.

$$M = \frac{eh}{4\pi m_e} = 9.27 \times 10^{-24} \text{ Am}^2$$

which is called as Bohr's magneton (μ_B).

~~My 1/24~~

Basic defⁿ $\rightarrow$

1. Magnetic flux (ϕ):-

Magnetic flux is the no. of magnetic field lines passing through a certain area.

2. Magnetic flux density (B):-

It is magnetic flux in the medium per unit area.

$$B = \frac{\phi}{A}$$

Unit of $B \Rightarrow \text{wb/m}^2$ or Tesla.

3. Magnetic field strength (H) →

The force experienced by unit north pole at any point in the magnetic field is called as magnetic field strength.

Unit of H $\Rightarrow$ A/m $\Rightarrow$ Ampere / metre.

4. Magnetization or Intensity of magnetization (I) -

- Magnetic dipole moment per unit volume is called as intensity of magnetization (I).

$$I = \frac{\vec{M}}{V} = \frac{i \times A}{m^3} = \frac{i}{m}$$

Unit of I $\Rightarrow$ Ampere / metre.

5. Magnetic susceptibility (χ) →

Ratio of magnetisation (M) to the magnetic field strength (H).

$$\chi = \frac{M}{H}$$

or

The magnetization produced in the material is directly proportional to applied magnetic strength.

$$\vec{I} \propto \vec{H}$$

$$I = \chi H$$

$$\chi = \frac{I}{H}$$

It is dimensionless.

6. Magnetic permeability (μ) →

The relation between $\vec{B}$, $\vec{H}$ & $\vec{I}$ is given by -

$$B = \mu_0 (\vec{H} + \vec{M})$$

where μ_0 = permeability of vacuum.

$$B = \mu_0 H \left(1 + \frac{M}{H} \right)$$

$$B = \mu_0 H (1 + \chi)$$

where $(1 + \chi) = \text{relative permeability} = \mu_r$

$$\vec{B} = \mu_0 \mu_r \vec{H} \quad \text{--- ①}$$

for air, $\mu_r = 1$

$$\vec{B} = \mu_0 \vec{H}$$

as μ_0 & μ_r are constant

$$\therefore \mu_0 \mu_r = \mu$$

where, $\mu = \text{permeability of material}$

$\therefore$ From eqⁿ ① -

$$B = \mu H$$

$\therefore \mu = \frac{B}{H}$

The permeability of material is defined as ratio of magnetic flux density to the magnetic field strength.

Classification of magnetism on the basis of permeability -

1) Diamagnetism:

- These materials have magnetic permeability (μ) is less than 1 ($\mu < 1$)
- The magnetic susceptibility is also less than one & negative.

- If magnetic field is applied to the diamagnetic diamagnet, the magnetization produced is in opposite direction i.e. magnetization is negative.
- Due to the magnetization produced in opposite direction, they weakly repel.
- Magnetic susceptibility is independent of temp.
- e.g. Bismuth, Antimony, Zn, Ag.

2) Paramagnetic material:

- For these material, the magnetic permeability is slightly greater than one ($\mu > 1$)
- The magnetic susceptibility (χ) is positive but less than one ($0 < \chi < 1$).
- When external magnetic field is applied, the weakly magnetization is produced in the same direction of applied magnetic field.
- Due to that, they weakly attract.
- The magnetic susceptibility of material is inversely proportional to the temp.

$$\chi = \frac{1}{T}$$

- e.g. iron, oxygen, etc.

3) Ferromagnetic substance:-

- The magnetic permeability for these material is very very greater than one ($\mu \gg \gg 1$)
- The magnetic susceptibility is also positive & greater than one ($\chi > 1$)
- If weak magnetic field is applied, then also the material possess strong magnetization in

the same direction of applied magnetic field.

- Due to that, they strongly attract.
- e.g. iron, cobalt, Nickel, etc.
- The magnetic susceptibility is inversely proportional to the temp.
- $\chi \propto \frac{1}{T - T_c} \rightarrow T_c = \text{Curie temperature}$

Application of magnetic device :=

1) Magnetic data storage :=

- In data storage system of computer, the data is ~~store~~ stored in the form of digital i.e. binary 0 or 1.
- Magnet has dipole. Each one can use for zero or one.
- The hard disk (magnetic disc) is made up of so many circular plate called as platter.
- Many platters are kept on the one common spindle.
- The surface is coated with magnetic iron oxide.
- The core of platter is made of aluminium. It is also coated with ferric oxide.
- Tracks := Each platter is divided into thousand of ~~circle~~ circles which have uniform & equal magnetization.
- Sector := Each track is subdivided into the smaller units called sectors. The actual data ~~store~~ stored in sectors.
- Read / Write head :=
- There is one read write head on the platter which moves to & fro motion.
- It is the interface between magnetic media & electronic component.

- When data is to be stored on hard disk, the head convert binary data to magnet pulse.
- When data is to be read, the head convert magnetic pulse to binary data.

2. Transformer core :=

- Magnetic core is a piece of magnetic material having high magnetic permeability use to increase the magnetic strength of transformer.
- specially ferromagnetic substance are used as a core of transformer.
- The use of magnetic material in the core of transformer increase the magnetic strength thousand times than the transformer without magnetic core.
- Magnetic material in the core of transformer increases the strength of magnetic field but it has also some disadvantages such as energy loss due to hysteresis & eddy current.

Material used as core :

① Amorphous steel :=

- These are made from several paper thin metallic tapes which reduces the flow of eddy current.
- These materials can used at high temp. easily.

② Amorphous Metals :=

- These type of materials have low conductivity so that eddy currents can be reduced.
- These can be used as a core of transformer for high magnetic strength.

③ Ferrite Ceramics :=

- These compounds are made of metal & iron oxide.

PAGE No.	
DATE	/ /

- These materials are serve as a insulator so that it can reduced the eddy current.

④ carbonyl iron core :=

- These materials are made of powdered carbonyl iron & gives stable performance for wide range of temp.
- These materials reduces the eddy current of high temp.